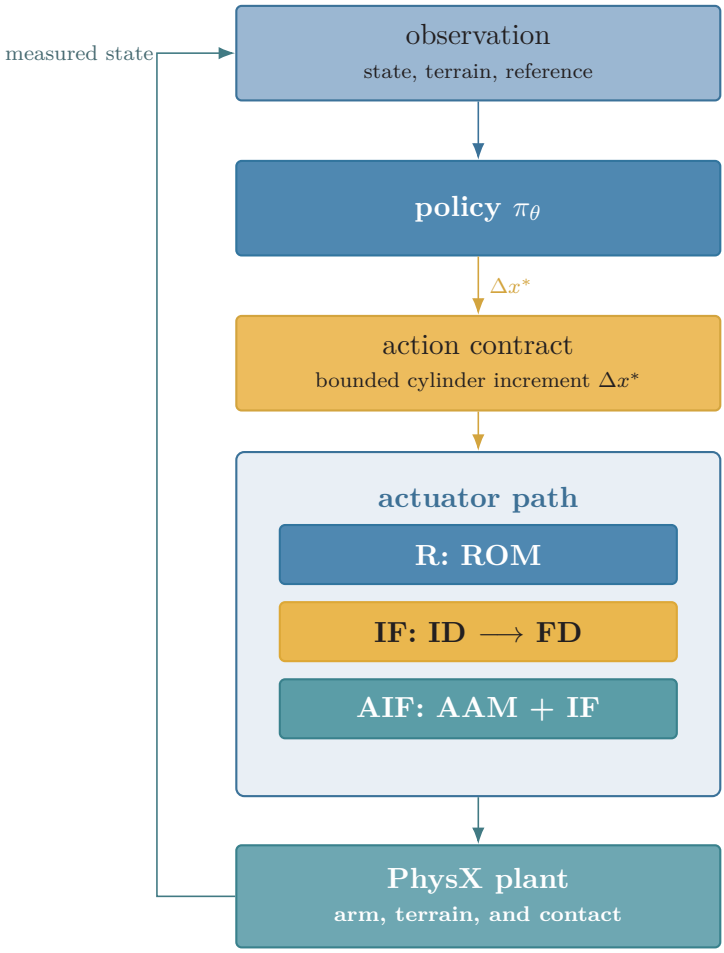


Figure 3.7: Deployed control stack. The policy produces a bounded cylinder increment; R, IF or AIF realizes it in the shared PhysX plant. Measured state closes the loop.

3.8 Problem Formulation

The reference is the verge centre line $y_{\text{ref}}(t)$ and cut height h_{ref}, expressed in the truck frame. With cutter-tip position $\boldsymbol{p}_c(t)$ and terrain height $g(\cdot)$, the scored errors are

$$e_{\text{lat}}(t) = p_{c,y}(t) - y_{\text{ref}}(t), \qquad e_{\text{cut}}(t) = \Big(p_{c,z}(t) - g(\boldsymbol{p}_c(t))\Big) - h_{\text{ref}}. \tag{3.7}$$

The objective is to minimize the mean and tail of both errors across path families and terrain while maintaining contact, head alignment, and smooth motion. Learned models are evaluated on one-step and recursive cylinder predictions. The stacks are tested across the randomized plant family; AIF is compared with IF on held-out factors.

Constraints are cylinder limits and headroom (Table 3.1), the action contract, checkpoint output limits, and the head-down condition.

3.9 System-Level Evaluation Criteria

Controllers and models are evaluated separately. Controller scores can hide actuator-model errors through policy compensation.